

Proposition 5.6 (Multifold sums of powers at zero binomial form). *For non-negative integers r, n, m*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (3.7), and Equation (1). □

Which, as well, can be expressed in terms of central factorial numbers of the second kind

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] k! T(k+1, m+1).$$

Thus, we have discussed quite remarkable results, that follow from centered Newton's formula evaluated at zero.